1.1) Valid distribution

All four values are between 0 and 1

$0.5600 + 0.1400 + 0.1200 + 0.1800 = 1.0000$

So the distribution is valid

1.2) p(dusty) and p(late)

$p(\text{dusty}) = 0.1200 + 0.1800 = 0.3000$

$p(\text{late}) = 0.1400 + 0.1800 = 0.3200$

1.3) p(late | dusty) and p(late | clear)

$p(\text{clear}) = 0.5600 + 0.1400 = 0.7000$

$p(\text{late} \mid \text{dusty}) = 0.1800 \div 0.3000 = 0.6000$

$p(\text{late} \mid \text{clear}) = 0.1400 \div 0.7000 = 0.2000$

1.4) $p(\text{dusty} \mid \text{late}) = 0.1800 \div 0.3200 = 0.5625$

Is it same as p(late | dusty)?

answer: It differs from 0.6000 because we ~~divide~~ divide by p(late) instead ~~of~~ p(dusty)

1.5) p(late | dusty) = 0.6000, while p(late) = 0.3200

The chance of being late changes when we know the weather, so they are dependent.

p(dusty) x p(late) = 0.3000 x 0.3200 = ~~0.0960~~

0.0960 ≠ 0.1800 = p(dusty, late), so they are not independent.

2.1) using the law of total probability

p(free) = (0.3000 x 0.6000) + (0.7000 x 0.0500)

= 0.1800 + 0.0350 = 0.2150

2.2) by Bayes' rule, p(spam | free) =

p(spam) x p(free | spam) ÷ p(free).

p(spam | free) = (0.3000 x 0.6000) ÷ 0.2150

= 0.8372

2.3) The words are independent once we ~~know~~ know the email type, so I multiply their chances.

p(spam and both words) = 0.3000 x 0.6000 x 0.4000 = 0.0720

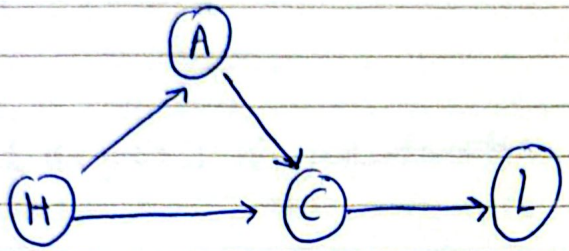
p(normal and both words) = 0.7000 x 0.0500 x 0.0200 = 0.0007

p(spam | free, winner) = 0.0720 ÷ (0.0720 + 0.0007) = 0.9904

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2.4) It went up from 0.8372 to 0.9904 because "winner" is more common in spam.

3.1)



H has no parents. A has H. C has H and A. L has C.

3.2) Joint probability

$$P(H, A, C, L) = P(H) \times P(A|H) \times P(C|H, A) \times P(L|C)$$

3.3) Count one "yes" probability for each combination of the parents' values. H needs 1, A needs 2, C needs 4, and L needs 2. Total = 1 + 2 + 4 + 2 = 9

3.4) $P(A = no \mid H = yes) = 1 - 0.6000 = 0.4000$

Joint probability = $0.3000 \times 0.4000 \times 0.2000 \times 0.8000 = 0.0192$.

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3.5) I add the two cases for A in each line

$P(H=yes, C=yes) = 0.3000 \times (0.6000 \times 0.7000 + 0.4000 \times 0.2000) = 0.1500$

$P(H=no, C=yes) = 0.7000 \times (0.1000 \times 0.5000 + 0.9000 \times 0.0500) = 0.0665$

$P(H=yes \mid C=yes) = 0.1500 \div (0.1500 + 0.0665) = 0.6928$

$P(H=yes \mid C=yes \mid L=yes)$ is also 0.6928.

L depends only on C, so once we know there was a power cut, the cancellation tells us nothing more about H.

4.1) The query is H. The evidence is L = yes.

The hidden variables are A and C.

Let $S(h, a) = P(H=h, A=a, L=yes)$. In each bracket, add the C=yes and C=no cases.

$S(yes, yes) = 0.3000 \times 0.6000 \times (0.7000 \times 0.8000 + 0.3000 \times 0.1000) = 0.1062$

$S(yes, no) = 0.3000 \times 0.4000 \times (0.2000 \times 0.8000 + 0.8000 \times 0.1000) = 0.0288$

$S(no, yes) = 0.7000 \times 0.1000 \times (0.5000 \times 0.8000 + 0.5000 \times 0.1000) = 0.0315$

$S(no, no) = 0.7000 \times 0.9000 \times (0.0500 \times 0.8000 + 0.9500 \times 0.1000) \approx 0.0851$

$P(H=yes, L=yes) = S(yes, yes) + S(yes, no) = 0.1350$

$P(H=no, L=yes) = S(no, yes) + S(no, no) \approx 0.1166$

$P(L=yes) = P(H=yes, L=yes) + P(H=no, L=yes) \approx 0.2516$

To normalize, divide by $P(L=yes)$ using its full value before rounding: $\alpha = 1 / P(L=yes)$

$P(H \mid L=yes) = \alpha \langle 0.1350, 0.1166 \rangle = \langle 0.5367, 0.4633 \rangle$

So $P(H=yes \mid L=yes) = P(H=yes, L=yes) \div P(L=yes) \approx 0.5367$

4.2) H and C are hidden. Using the sums above, keep the two cases with A=yes.

$P(A=yes, L=yes) = S(yes, yes) + S(no, yes) =$

$0.1062 + 0.0315 = 0.1377$

$P(A=yes \mid L=yes) = 0.1377 \div P(L=yes) \approx$

0.5474

4.3) Only C is hidden. The two cases with A=no give the new total.

$P(A=no, L=yes) = S(yes, no) + S(no, no) \approx 0.1139$

$P(H=yes \mid L=yes, A=no) = S(yes, no) \div P(A=no, L=yes)$

≈ 0.2530

It went down from 0.5367 to 0.2530.

Heatwaves make overloads more likely, so finding no overload makes a heatwave less likely. A heatwave can still cause a power cut directly through H→C, without an overload, a power cut is much less likely, so the cancellation is weaker evidence for a heatwave.

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5a) 1. Keep only the rows with L=yes

Keep 2, 4, 6, 7, 12. Throw away 1, 3, 5, 8, 9, 10, 11.

Only rows 2 and 7 of the five kept rows have H = yes.

$p(H = yes \mid L = yes) \approx 2 \div 5 = 0.4000$

2. Fraction thrown away

7 out of 12 were thrown away, so the fraction is $7/12 \approx 0.5833$.

When the evidence is rare, most samples get thrown away and very few are left to make the estimate.

5 b)

1. The weight is $p(L = yes \mid C)$. Use 0.8000 if C = yes and 0.1000 if C = no.

Rows 1 to 6 have weight 0.1000, 0.8000, 0.1000, 0.1000, 0.8000, 0.8000.

2. Weight for H = yes (rows 2, 4, 6) = 0.8000 + 0.1000 + 0.8000 = 1.7000

All weights = 0.1000 + 0.8000 + 0.1000 + 0.1000 + 0.8000 + 0.8000 = 2.7000

$p(H = yes \mid L = yes) \approx 1.7000 \div 2.7000 \approx 0.6296$

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3. Rejection sampling gave 0.4000 and likelihood weighting gave 0.6296. The exact answer was 0.5367. Both estimates used only a few random samples, so the results can vary. Using more samples usually gets them closer.

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